

ThousandOfWind at GroundLM 2026 Shared Task: An Auditable Hybrid Pipeline for LitTraceQA

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Abstract

We describe the ThousandOfWind system for LitTraceQA, the GroundLM 2026 shared task on evidence-grounded question answering over scientific literature. The system retrieves papers from a 27,487-document metadata pool, localizes supporting pages and document objects, and produces multiple-choice, free-form, and schema-constrained table answers. It combines word and character TF-IDF, BGE-M3 dense retrieval, query-derived source planning, PDF page and object ranking, deterministic structural extractors, and guarded Qwen fallbacks. On the 55-query public development set, our final audited configuration obtains 0.720 Paper F1, 0.551 Evidence F1, 0.902 multiple-choice accuracy, 0.846 free-form exact match, and 0.791 table row F1. The submitted input-only run covers all 71 test queries and passes the released schema validator. Our central finding is methodological: aggressive development-set heuristics can produce misleading gains, so we retain only behavior supported by type-level rules, multi-document probes, and label-isolated execution. The final system was selected through a versioned optimization campaign that preserves both successful and rejected ablations.

1 Introduction

Scientific question answering is useful only when a reader can trace an answer to its sources. LitTraceQA (Liu et al., 2026) evaluates this complete chain: retrieve one or more relevant papers, locate coarse evidence such as a page, figure, equation, or citation context, and return an answer in the requested format. The task includes single- and multi-paper questions and evaluates retrieval, grounding, and answer correctness separately.

Our system addresses three practical difficulties. First, questions refer to papers through heterogeneous signals, including titles, acronyms, methods, venues, datasets, and experimental conditions. Second, scientific PDFs flatten tables and figures in

ways that make text-only localization unreliable. Third, a small public development set makes repeated manual optimization especially vulnerable to overfitting.

We therefore build an auditable hybrid pipeline with four stages: (1) lexical and dense paper retrieval, (2) query-derived source-cardinality planning, (3) manifestation-aware PDF evidence localization, and (4) deterministic answer extraction followed by guarded generation. All runtime decisions use only the question, released answer metadata, paper metadata, and retrieved PDF content. Gold papers, evidence, and answers are used only by the evaluator.

Our development process is also part of the contribution. Each experiment stores predictions, aggregate metrics, per-query results, parameters, and model provenance. Candidate rules must express a reusable question or document pattern, work on at least two documents, and pass a self-constructed probe. We removed earlier rules that named individual benchmarks, methods, object numbers, or expected answer patterns, even when doing so lowered public scores.

2 Task and Resources

Each input contains a stable query identifier, a natural-language question, requested answer types, and public answer metadata. Multiple-choice questions expose their option labels and text; table questions expose an ordered schema with column types and row-key indicators. A prediction contains paper identifiers, evidence locations, and an answer object. The public development set has 55 labeled queries; the required test split has 71 input-only queries.

The official evaluator reports macro paper precision, recall, and F1; macro evidence precision, recall, and F1; multiple-choice accuracy; free-form exact match; and table row and cell metrics. We

use the released metadata pool and PDFs obtainable from its public links. External components are scikit-learn, BGE-M3 (Chen et al., 2024), Qwen3 (Yang et al., 2025), and Qwen3-VL. All model inference is local through Ollama; we use no additional training data and perform no fine-tuning.

3 System

3.1 Paper Retrieval

We index concatenated title, abstract, author, venue, and year fields with a word-level TF-IDF vectorizer. Separate title and character 3–5-gram indexes improve robustness to stylized titles, acronym segmentation, and OCR artifacts. A BGE-M3 embedding index contributes a dense cosine score with weight 0.2. The final score is a weighted sum of these signals and corpus-derived title aliases.

Retrieval depth is inferred from visible query structure. The default depth is one paper for an explicit single source and four for a multi-source query; venue-wide surveys may retrieve up to twelve. For questions that visibly name two independent source entities, the system can retrieve each entity separately and fuse the cohorts. This path is accepted only when complete answers synthesized from the baseline and fused cohorts agree exactly, preventing retrieval expansion from silently changing an otherwise stable answer.

3.2 Evidence Grounding

For each selected paper, we cache both plain and layout-preserving PDF text. Pages are initially ranked through lexical retrieval over page pseudo-documents. The ranker then applies source-aware constraints inferred from the question. Explicit “Table n ” or “Figure n ” mentions require that object label in the page text; implicit table and figure questions favor caption-bearing pages; citation questions search numbered reference entries; and equation questions score displayed equation markers.

The localizer emits at most five evidence records and normally selects one page per paper. Structured locators are derived from visible labels and document content rather than fixed table or figure numbers. The answer synthesizer receives a broader context of up to four pages per paper, so evidence-output precision is not coupled to the context budget.

3.3 Answer Synthesis

The answer layer first applies deterministic extractors when the requested value can be bound to document structure. These cover reference entries, hardware configurations, displayed equations, absolute table deltas, repeated metric columns, method-value mappings, and document-defined configuration markers. The extractors parse entities and constraints from the question and bind them to PDF captions, headers, rows, or nearby prose.

Remaining multiple-choice and free-form questions use Qwen3 with retrieved page text. Generation is guarded by option alignment and structural post-processing. A Qwen3-VL fallback is restricted to figure questions for which visual evidence is necessary.

Table answers are generated against the released schema and then sanitized. Every row must contain exactly the schema columns; numeric and Boolean values are converted to their JSON types; missing non-key cells become null; and rows without required keys are removed. Grounding uses more candidate evidence than the final evidence array to reduce row-to-paper binding failures.

3.4 Label Isolation

The same command-line pipeline processes labeled development inputs and input-only test inputs. With `-skip-evaluation`, the gold file is never read, manifests record no gold path, and query artifacts omit gold sections. We exercised this path with a nonexistent gold filename. The final test artifact contains 71 unique query identifiers and passes exact-coverage and answer-schema validation.

4 Experiments

4.1 Setup

We evaluate with the released local evaluator. The accepted configuration uses semantic weight 0.2, one/four papers for single-/multi-source questions, one localized page per selected paper, at most five returned evidence records, and four synthesis pages per paper. Qwen3 is used for guarded answer and table fallback, while Qwen3-VL is used only by the figure guard.

Table 1 reports the current audited system. Paper retrieval has high precision (0.895) but lower recall (0.677), which limits Paper F1. Evidence localization remains the largest pipeline bottleneck.

Metric	Score
Paper precision / recall / F1	0.895 / 0.677 / 0.720
Evidence precision / recall / F1	0.581 / 0.541 / 0.551
Multiple-choice accuracy	0.902
Free-form exact match	0.846
Table row F1	0.791
Table cell accuracy (macro)	0.722
Table cell accuracy (micro)	0.778

Table 1: Public development results of the final audited, label-isolated runtime.

System	Paper	Evid.	MC	Cell-M
Audited base	.636	.517	.780	.609
Final audited	.720	.551	.902	.722

Table 2: Development F1/accuracy across accepted audited configurations. Cell-M is macro table cell accuracy.

4.2 Optimization Campaign

Table 2 compares the first repository-wide audited baseline with the final system. Subsequent accepted changes use generic table structure, query decomposition, and answer-stable two-entity composition. They improve paper retrieval, multiple-choice accuracy, and table cell accuracy without changing free-form exact match or table row F1.

We organized optimization as a four-stage loop: analysis, research, ablation, and review. At the start of each round, we compared all seven standard metrics and inspected per-query retrieval, locators, answers, prompts, and cached PDF text. We grouped failures into missing-paper, wrong-rank, wrong-page/object, table binding, visual grounding, and answer-normalization classes. Candidate techniques were then assessed for expected impact, runtime cost, reproducibility, and leakage risk.

Every behavior-changing round used a fresh branch, an immutable control run, a full candidate, and a factor-removal variant when feasible. The generalization gate required a type-level claim, evidence from at least two documents, and a self-constructed probe. Runtime inputs were checked explicitly to exclude gold answers, paper IDs, evidence, table rows, validation labels, and query-ID routes. An independent review inspected correctness and development-set overfitting before a merge decision. Accepted runtime changes and failed experiments followed different paths: winners were merged through normal pull requests, while rejected or inconclusive runs were preserved in artifacts-only pull requests with no active behav-

ior. A generated dashboard tracked one comparable winner per round while retaining all variants for diagnosis.

This discipline changed the final system. Answer-stable two-entity composition improved Paper F1 by 0.015 without changing any public answer, while later cleanup rounds verified byte-identical public and test predictions after rejected verifier experiments. Negative experiments were equally informative: global layout-text replacement damaged table alignment; unrestricted answer-aware page reranking promoted numeric distractors; reciprocal-rank fusion reduced several answer metrics; and raw grid injection collapsed correct tables.

An earlier phase obtained substantially higher public scores by accumulating narrow named rules. The audit found validation-shaped method, benchmark, and object assumptions, so we removed them and do not report that configuration as the submitted system. Thus the final score reflects the best reviewed runtime, not the maximum observed development score.

4.3 Test Submission

Our final prediction file contains all 71 required test queries, with 50 multiple-choice and 21 table answer objects. Four records have empty evidence arrays; these remain schema-valid but are a known grounding risk. Official hidden-label scores were not available when this paper was rendered, so we make no test-accuracy claim. The camera-ready version will distinguish official submitted scores from any later analysis.

5 Error Analysis

Cross-paper cardinality. Questions can name several methods without stating whether their values occur in one comparison paper or in separate source papers. Conservative source planning improves precision but misses implicit companion papers, explaining much of the paper-recall gap.

PDF manifestations and page namespaces. Public links can resolve to versions whose pagination or supplementary-file layout differs from the annotation source. Semantically correct content may therefore appear at a different physical page. We do not emit alternative page numbers unless they are derivable from the retrieved manifestation.

Tables and figures. PDF text extraction can merge headers, rows, and adjacent columns. Table

errors are more often incorrect row-to-paper or condition binding than invalid JSON. Figures pose the complementary problem: captions are searchable, but panel-level visual details may not survive text extraction.

Answer support. A correct option can be selected from weak or unrelated evidence. Our audits therefore inspect source support separately from answer accuracy. This distinction is important because multiple-choice accuracy is considerably higher than Evidence F1.

6 Reproducibility

The system is implemented in Python and released at <https://github.com/ThousandOfWind/LitTraceQA>. Run manifests record model names and digests, Ollama version, parameters, input paths, cache provenance, and output paths. The test command is:

```
python scripts/run_baseline_eval.py
-input data/littraceqa/test.jsonl
-gold /nonexistent/hidden.jsonl
-skip-evaluation
```

The repository includes the released evaluator and validator, 226 unit tests, immutable run directories, per-query comparison artifacts, and ablation summaries. The submitted prediction is identified by its SHA-256 digest in the paper metadata.

7 Conclusion

We presented an auditable hybrid system for LitTraceQA. Lexical and dense retrieval, query-derived source planning, manifestation-aware PDF localization, structural extraction, and guarded local generation yield strong answer accuracy while exposing a persistent evidence-grounding gap. More importantly, strict label isolation and de-overfitting audits changed which development gains we considered valid. Future improvements should focus on learned scientific reranking and layout-aware page/object retrieval evaluated on held-out documents.

8 Limitations

The development set contains only 55 queries, so metric estimates have high variance and can reward narrow rules. Although our final audit removes known named validation routes, it cannot prove generalization to every unseen document

style. PDF extraction is sensitive to publisher layout, OCR quality, versioning, and supplementary-page conventions. Local Qwen outputs can vary unless model digests and cached responses are fixed. The system does not verify the scientific truth of a paper’s claims; an evidence locator indicates textual support, not independent validation. Finally, official test scores are pending, and validator success demonstrates format correctness rather than answer correctness.

Ethics Statement

The system processes publicly linked scientific papers and does not use personal user data. Paper metadata and PDFs remain subject to their original licenses; our repository stores derived caches locally rather than redistributing PDFs. Generated answers and evidence should be checked by a reader before being used in research or decision making.

References

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